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Recognition of Acoustic Signals of Commutator Motors

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Abstract: Most faults can stop a motor, and time is lost in fixing the damaged motor. This is a reason why it is essential to develop fault-detection methods. This paper describes the acoustic-based fault detection of two commutator motors: the commutator motor of an electric impact drill and the commutator motor of a blender. Acoustic signals were recorded by a smartphone. Five states of the electric impact drill and three states of the blender were analysed: for the electric impact drill, these states were healthy, damaged gear train, faulty fan with five broken rotor blades, faulty fan with 10 broken rotor blades, and shifted brush (motor off); for the blender, these states were healthy, faulty fan with two broken rotor blades, and faulty fan with five broken rotor blades. A feature extraction method, MSAF-RATIO-27-MULTIEXPANDED-4-GROUPS (Method of Selection of Amplitudes of Frequency Ratio of 27% Multiexpanded 4 Groups), was developed and used for the computation of feature vectors. The nearest mean (NM) and support vector machine (SVM) classifiers were used for data classification. Analysis of the recognition of acoustic signals was carried out. The analysed value of TE_{EID} (the total efficiency of recognition of the electric impact drill) was equal to 96% for the NM classifier and 88.8% for SVM. The analysed value of TE_B (the total efficiency of recognition of the blender) was equal to 100% for the NM classifier and 94.11% for SVM.

Keywords: electrical machine; classification; mechanical fault; detection; analysis; sound; commutator motors

1. Introduction

Commutator motors (CMs) are the most frequently used electrical motors for devices used at home, for example, electric impact drills, angle grinders, cordless drills, hair dryers, blenders, etc. CMs are also used in the energy industry, material industry, construction industry, mining industry, transport industry, and by construction and renovation companies. Sometimes, faults of devices can happen. An undetected fault can lead to the damage of the whole device. Frequent faults of commutator motors are: bearing faults, damaged gears, damaged teeth, short-circuits, and damaged electric motor brushes. Bearings hold the shaft of the CM, but they wear out after years of use. Most of these faults can stop the motor. Then, there is a loss of time associated with fixing the damaged motor. It is essential for industrial motors. A production line (consisting of motors) can be stopped for a longer time for repair.

The above constitutes an essential reason for developing fault-detection methods. Diagnostic methods can use various signals, for example: ultrasound, acoustic signals, current signals, magnetic signals, vibration, video images, and thermal images. There is also the possibility of using oil analysis. Ultrasound is used for detecting faults of motors, pumps, and bearings [1–4]. However, the cost of an ultrasound generator is about $1000. The analysis of acoustic emissions (in the range 30–16,000 Hz) is noninvasive and low-cost [5–12]; however, its main disadvantages are the appearance

of background noises and reflected sounds. The analysis of thermal images is also a common diagnostic technique [13–15], which is used for electrical faults such as shorted coils and broken bars of the motor. The analysis of electric currents is also used for the fault diagnosis of motors [16–21]. Similarly to the analysis of thermal images, it can be used for electrical faults. Vibration analysis is often used for the condition monitoring of motors [22–28]. Vibration analysis can detect many types of faults, such as rotor and stator faults, mechanical faults, gear faults, cracks, bearing faults, damaged teeth of sprockets, etc. Moreover, a vibration signal is measured immediately.

This paper deals with the fault detection of an electric impact drill and blender using a smartphone. Considering that the electric impact drill and blender consisted of commutator motors and other mechanical parts such as gears, sprockets, etc., five states of the electric impact drill were analysed: healthy (Figure 1), with a damaged gear train (Figure 2), with a faulty fan with five broken rotor blades (Figure 3), with a faulty fan with 10 broken rotor blades (Figure 4), and with a shifted brush (motor off) (Figure 5); and three states of the blender were analysed: healthy (Figure 6), with a faulty fan with 2 broken rotor blades (Figure 7), and with a faulty fan with 5 broken rotor blades (Figure 8).

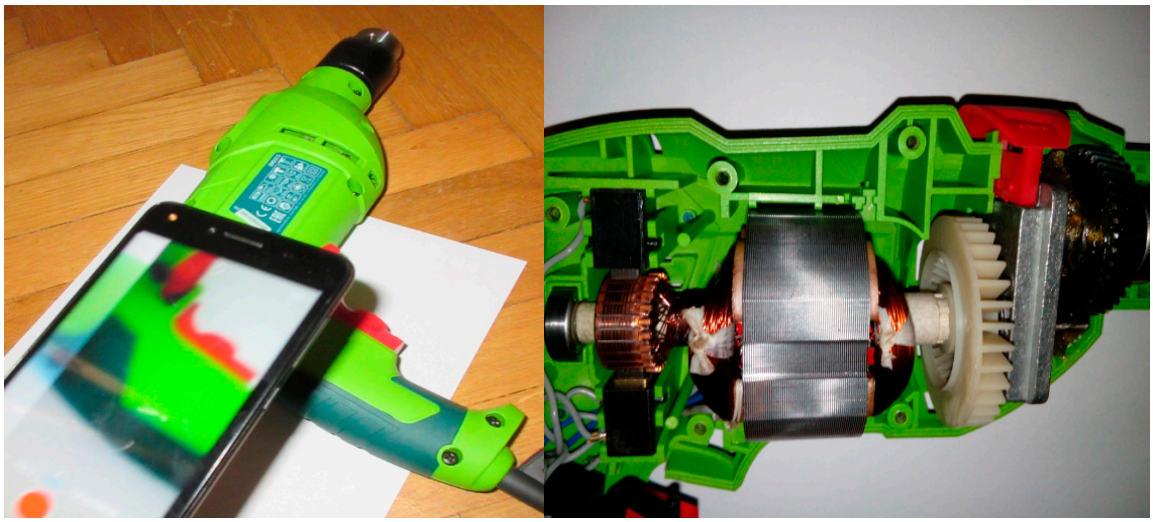


Figure 1. Healthy electric impact drill with smartphone.

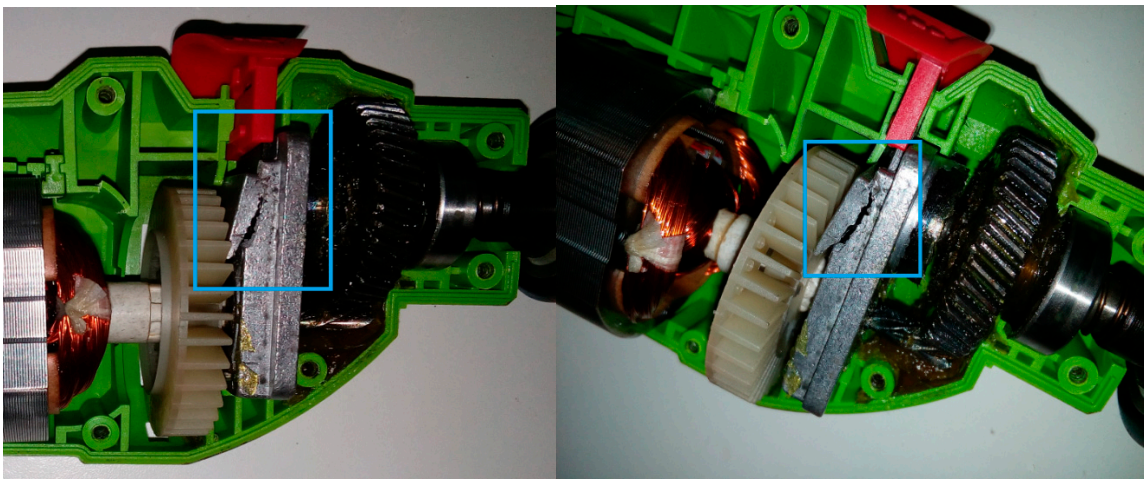


Figure 2. Electric impact drill with a damaged gear train (indicated by the blue box).

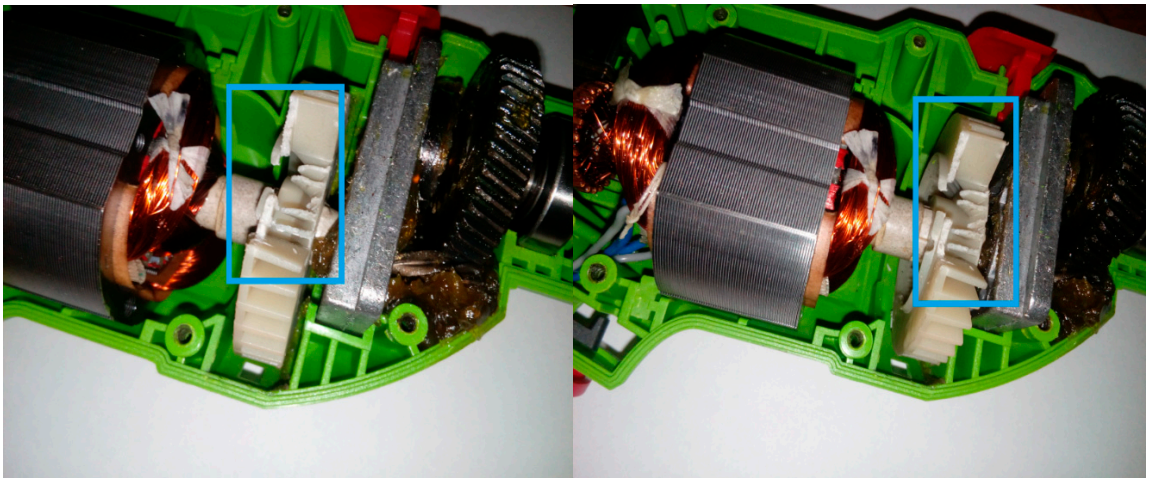


Figure 3. Electric impact drill with a faulty fan (5 broken rotor blades; indicated by the blue box).

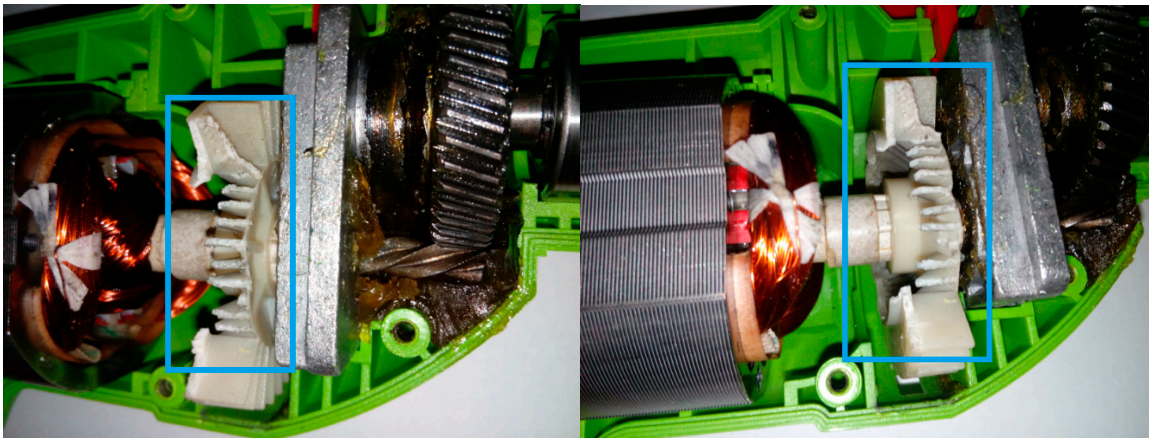


Figure 4. Electric impact drill with a faulty fan (10 broken rotor blades; indicated by the blue box).

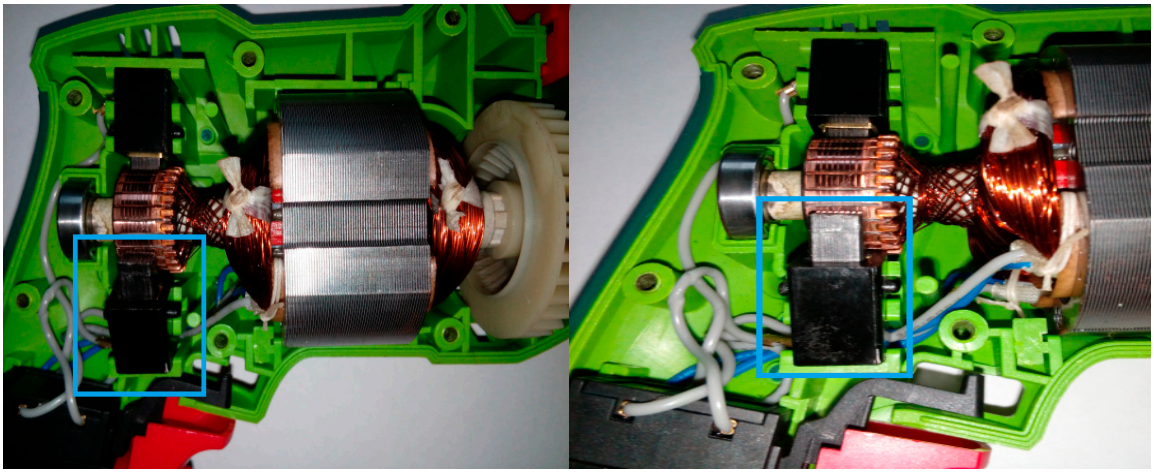


Figure 5. Electric impact drill with a shifted brush (motor off; indicated by the blue box).



Figure 6. Healthy blender.

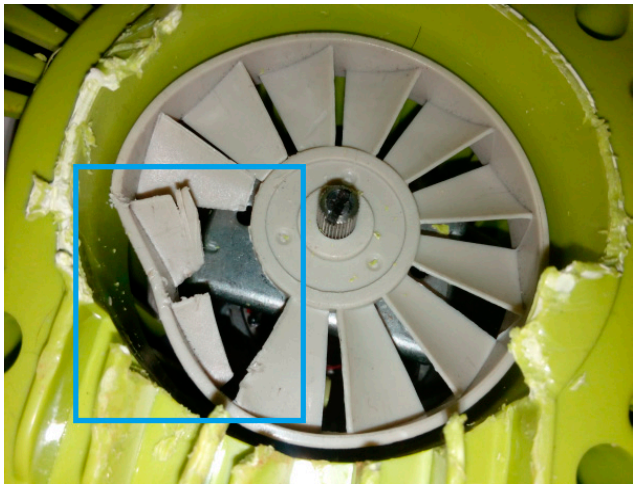


Figure 7. Blender with a faulty fan (2 broken rotor blades; indicated by the blue box).

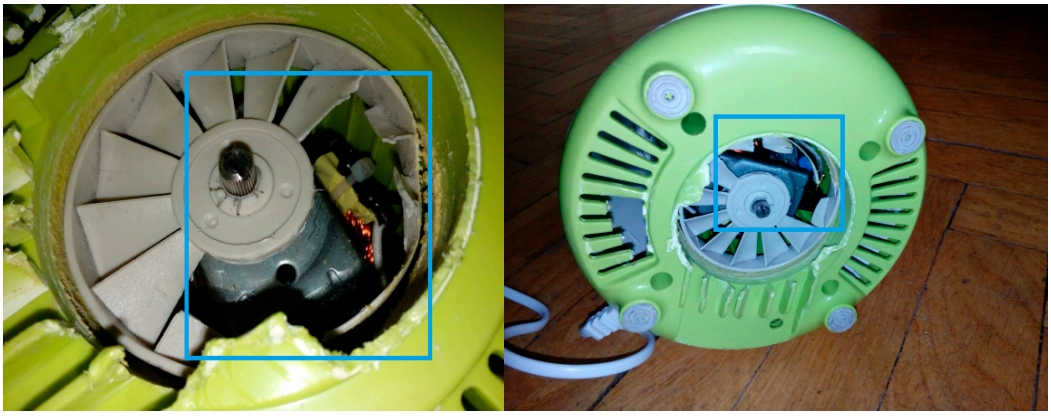


Figure 8. Blender with a faulty fan (5 broken rotor blades; indicated by the blue box).

The developed approach used a smartphone as the measuring device. Acoustic signals were recorded by the smartphone. Next, the signals were processed by a method of amplitude normalization, MSAF-RATIO-27-MULTIEXPANDED-4-GROUPS (Method of Selection of Amplitudes of Frequency Ratio 27% Multiexpanded 4 Groups—described more in detail in Section 2.1) and nearest mean and SVM classifiers. The results of the fault detection for the electric impact drill and blender are presented in Section 3.

2. Proposed Fault Detection Approach Using a Smartphone and Acoustic Signals

The smartphone can record acoustic signals (parameters of a microphone: frequency response: 30–16,000 Hz, sensitivity: −42 dB). The smartphone was positioned above the electric impact drill/blender (0.4 m above the device). Next, the recorded acoustic signals were converted into acoustic data using 'MPlayer library—The Movie Player'. Obtained acoustic data (single channel, 44.1-kHz sampling frequency) were processed by signal processing methods: splitting the soundtrack into 1-s samples, a method of amplitude normalization, MSAF-RATIO-27-MULTIEXPANDED-4-GROUPS, the nearest mean classifier, and the SVM classifier. A flowchart of the proposed fault-detection approach is presented in Figure 9. An experimental setup of the measurements using the smartphone is presented in Figure 10a,b.

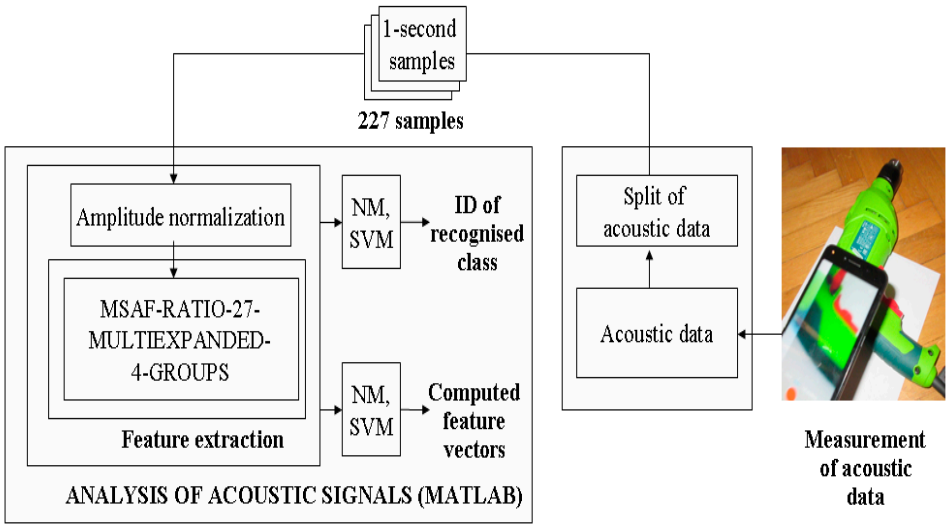
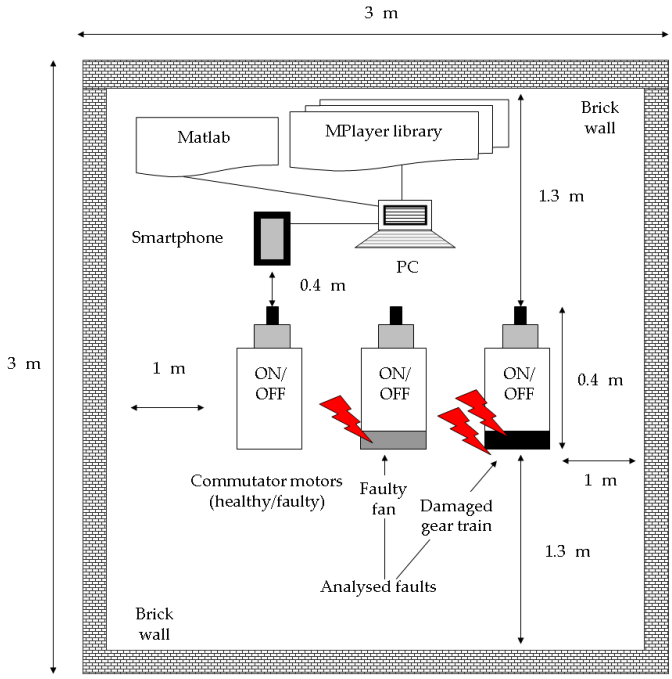
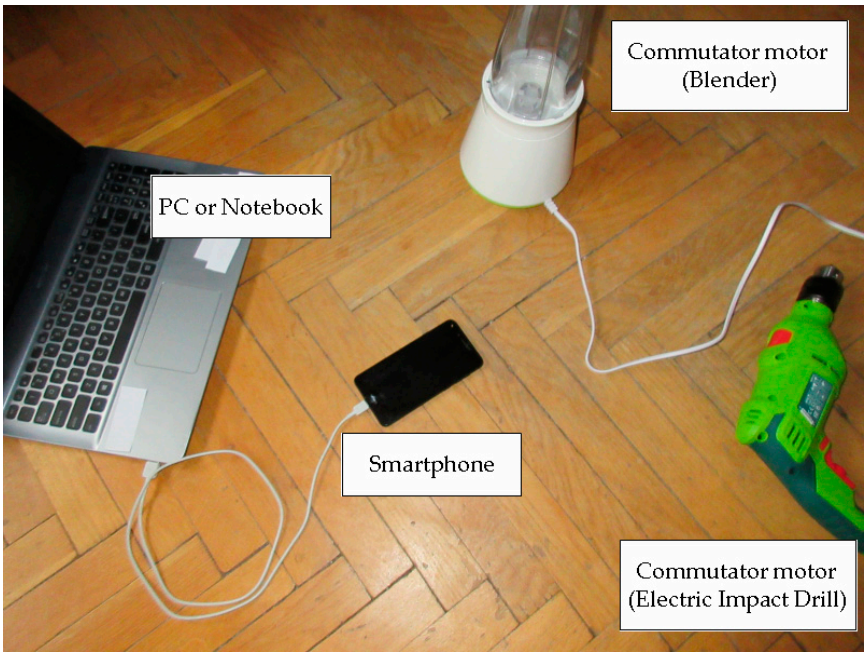


Figure 9. Flowchart of the proposed fault detection using smartphone and acoustic signals. NM: nearest mean; SVM: support vector machine.



(**a**)

Figure 10. *Cont.*



(**b**)

Figure 10. (**a**) Diagram of the experimental setup of the measurements using the smartphone. (**b**) Experimental setup of the blender and electric impact drill.

2.1. MSAF-RATIO-27-MULTIEXPANDED-4-GROUPS

It is difficult to see changes of signals in the time domain; this is a reason to use the frequency domain. The MSAF-RATIO-27-MULTIEXPANDED-4-GROUPS (Method of Selection of Amplitudes of Frequency Ratio 27% Multiexpanded 4 Groups) method computed the FFT (Fast Fourier Transform) spectra. Next, the method computed feature vectors of the analysed states of the electric impact drill. The method used the differences of the FFT spectra. The proposed method used the following steps:

1. Compute the FFT spectra of the analysed states of the electric impact drill. Each FFT spectrum had 16,384 values. These values formed 16,384 elements of the vector. Vectors were denoted as follows: healthy electric impact drill—**hd** = $[hd_1, hd_2, \ldots, hd_{16,384}]$, electric impact drill with damaged gear train—**ddgt** = $[ddgt_1, ddgt_2, \ldots, ddgt_{16,384}]$, electric impact drill with a faulty fan (5 broken rotor blades)—**dfive** = $[dfive_1, dfive_2, \ldots, dfive_{16,384}]$, electric impact drill with a faulty fan (10 broken rotor blades)—**dten** = $[dten_1, dten_2, \ldots, dten_{16,384}]$,
2. Find difference between the FFT spectra: **hd** − **ddgt**, **hd** − **dfive**, **hd** − **dten**, **ddgt** − **dfive**, **ddgt** − **dten**, **dfive** − **dten**.
3. Compute absolute values: |**hd** − **ddgt**|, |**hd** − **dfive**|, |**hd** − **dten**|, |**ddgt** − **dfive**|, |**ddgt** − **dten**|, |**dfive** − **dten**|.
4. Compute *Ratio* from Equation (1) for all absolute values:

$$Ratio = (100\%)AV_i/MaxAV \quad (1)$$

where *MaxAV* is the maximum absolute value of the difference between FFT spectra, AV_i is the absolute value of the difference between FFT spectra with index *i*, and *Ratio* is the threshold of the difference between FFT spectra. The RATIO-27 means that 27% of the maximum absolute value of the difference between FFT spectra was the threshold. First, the ratio is defined, and next, it is fixed to a specific threshold. All frequency components larger than the threshold were used for the further steps of processing.

5. Compute common frequency components using the parameter *TCF* (*Threshold of common frequency components*) = (*number of required common frequency components)/(number of all differences*). An example of using *TCF* is presented as follows. There are the following differences of the FFT spectra: |**hd** − **ddgt**|, |**hd** − **dfive**|, |**hd** − **dten**|, |**ddgt** − **dfive**|, |**ddgt** − **dten**|, |**dfive** − **dten**|. Frequency components of 100 and 150 Hz are selected 3 times for |**hd** − **ddgt**|. Frequency components of 100 and 200 Hz are selected 3 times for |**hd** − **dfive**|. Frequency components of 100 and 300 Hz are selected 3 times for |**hd** − **dten**|. Frequency components of 400 and 450 Hz are selected 3 times for |**ddgt** − **dfive**|. Frequency components of 400 and 500 Hz are selected 3 times for |**ddgt** − **dten**|. Frequency components of 400 and 600 Hz are selected 3 times for |**dfive** − **dten**|. It can be noticed that the best frequency components are 100 and 400 Hz, being found 9 times (if *TCF* = 9/18 = 0.5, it assumes, that 9 required common frequency components are set and there are 18 differences between 4 states, it assumes there are 3 training sets, then it will have 18 differences). The MSAF-RATIO-27-MULTIEXPANDED-4-GROUPS finds none of the frequency components (if *TCF* = 11/18 = 0.6111 − 11 required common frequency components/18 differences).
6. Group the common frequency components. The MSAF-RATIO-27-MULTIEXPANDED-4-GROUPS used 1–4 groups of the best common frequency components. In the mentioned example, the best group had the common frequency components of 100 and 400 Hz.
7. Use 1–4 of the found groups.
8. Form a feature vector.

A flowchart of the proposed method of feature extraction, MSAF-RATIO-27-MULTIEXPANDED-4-GROUPS, is shown in Figure 11.

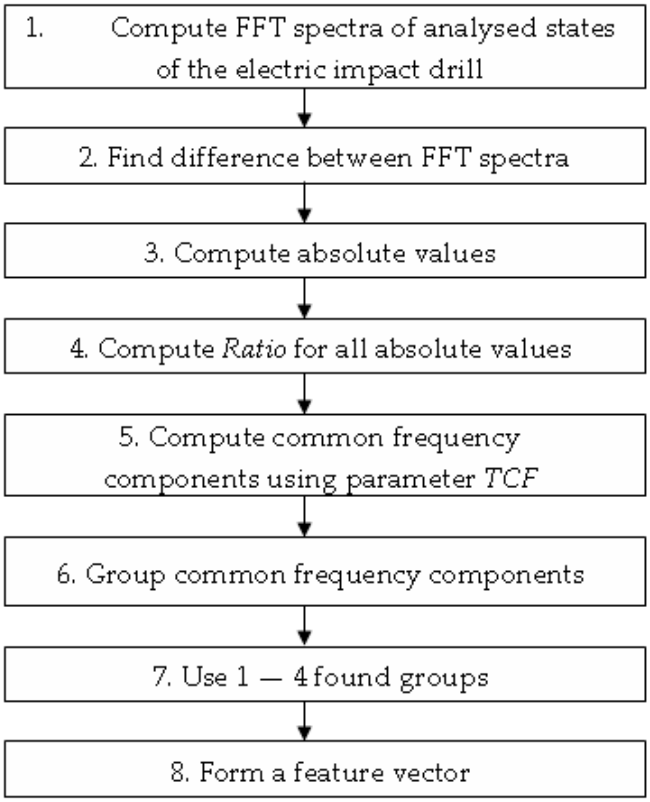


Figure 11. Flowchart of the proposed method of feature extraction, MSAF-RATIO-27-MULTIEXPANDED-4-GROUPS. FFT: Fast Fourier Transform; TCF: threshold of common frequency components.

Computed differences between the FFT spectra of the analysed states of the electric impact drill (rotor speed: 3000 rpm) are shown in Figures 12–17. Next, we can select the best frequency components for all differences.

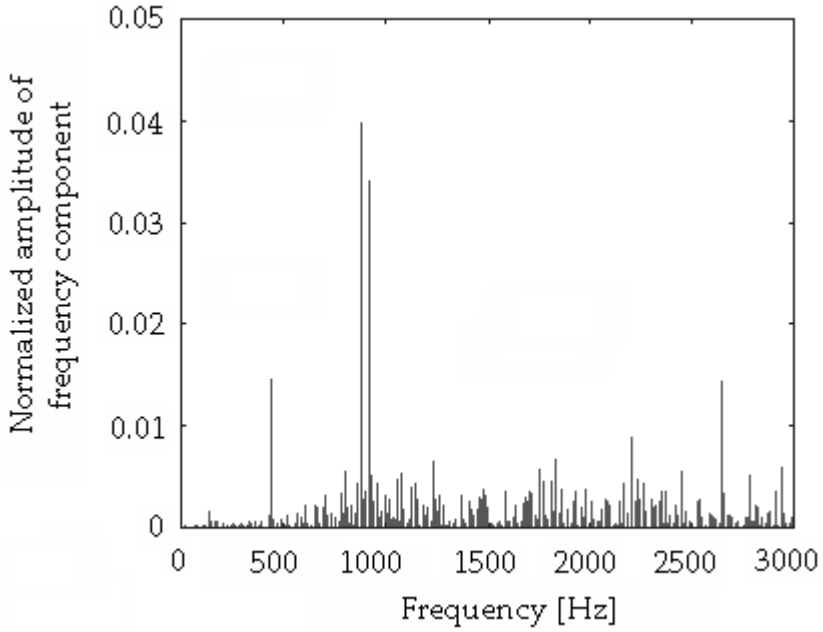


Figure 12. Difference (|**hd** − **ddgt**|) using the MSAF-RATIO-27-MULTIEXPANDED-4-GROUPS method. **hd** refers to the healthy electric impact drill; **ddgt** refers to the electric impact drill with damaged gear train.

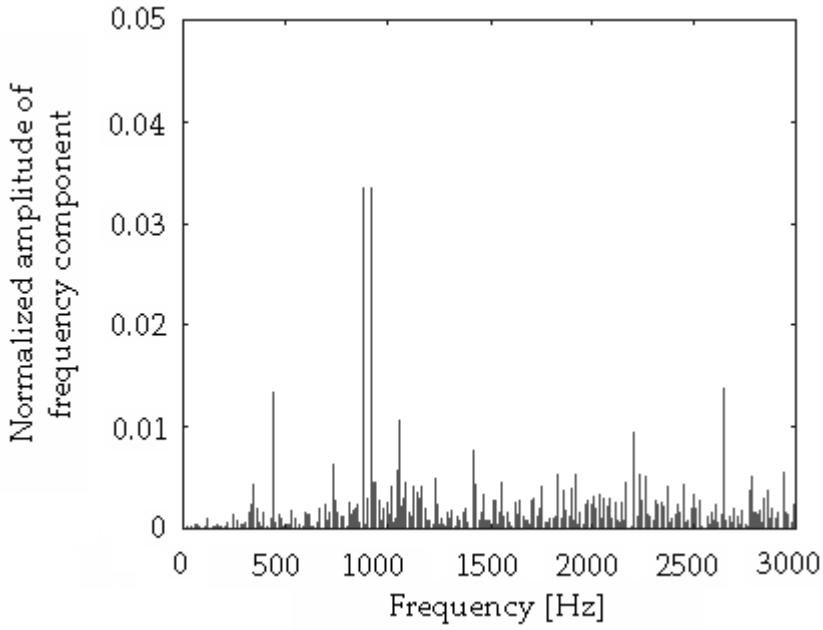


Figure 13. Difference (|**hd** − **dfive**|) using the MSAF-RATIO-27-MULTIEXPANDED-4-GROUPS method. **dfive** refers to the electric impact drill with a faulty fan (5 broken rotor blades).

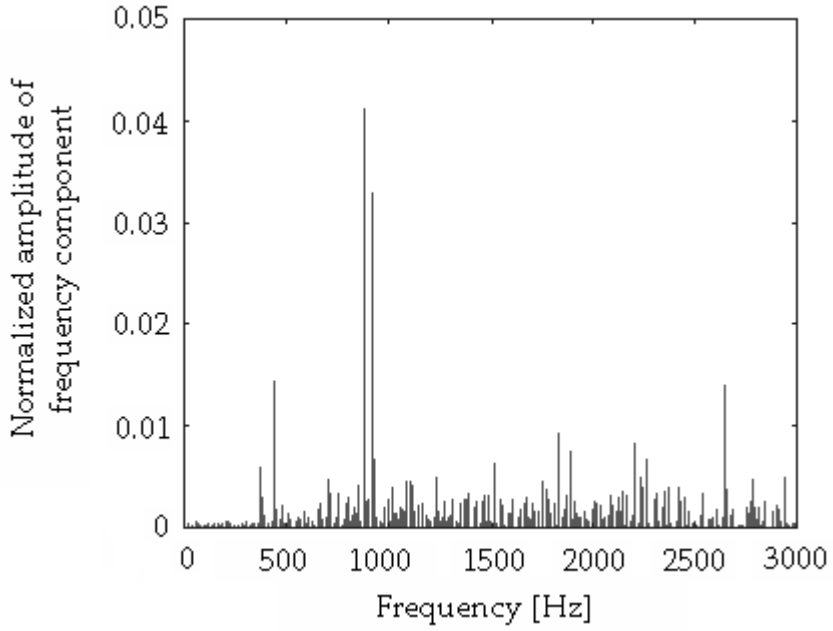


Figure 14. Difference ($|\mathbf{hd} - \mathbf{dten}|$) using the MSAF-RATIO-27-MULTIEXPANDED-4-GROUPS method. **dten** refers to the electric impact drill with a faulty fan (10 broken rotor blades).

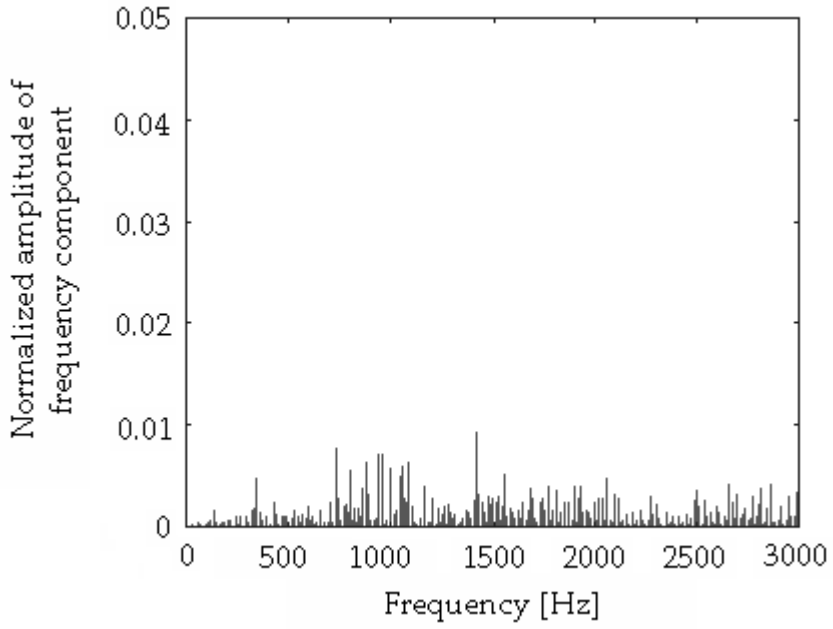


Figure 15. Difference ($|\mathbf{ddgt} - \mathbf{dfive}|$) using the MSAF-RATIO-27-MULTIEXPANDED-4-GROUPS method.

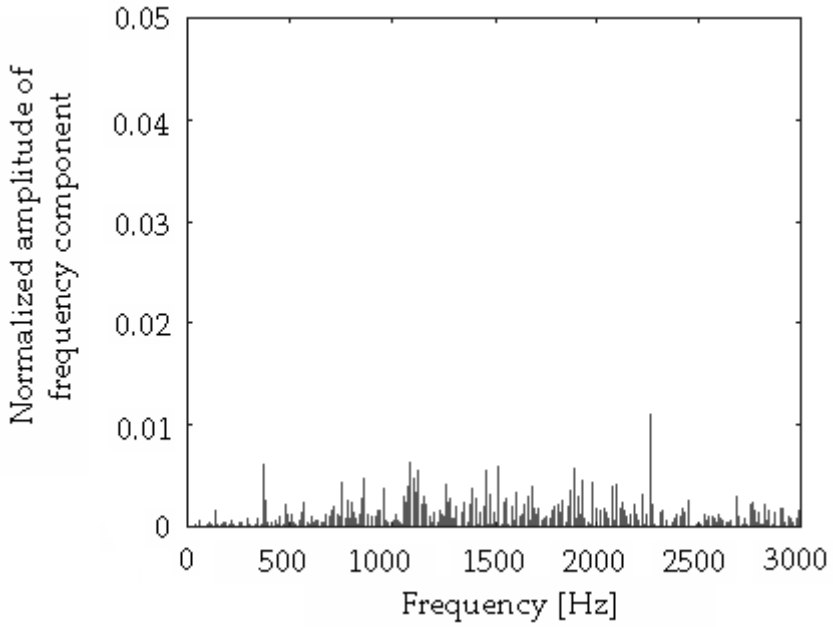


Figure 16. Difference (| **ddgt** − **dten** |) using the MSAF-RATIO-27-MULTIEXPANDED-4-GROUPS method.

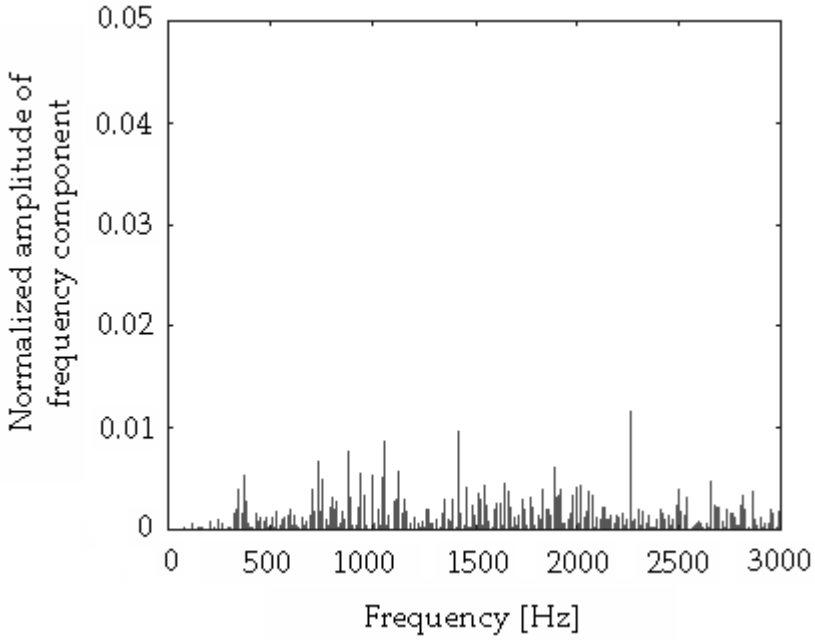


Figure 17. Difference (| **dfive** − **dten** |) using the MSAF-RATIO-27-MULTIEXPANDED-4-GROUPS method.

The analysed states are compared with each other. If there are 3 states, 3 differences are analysed for 1 training set. If there are 4 states, 6 differences are analysed for 1 training set. If there are 5 states, 10 differences are analysed for 1 training set. If there are 5 states, 30 differences are analysed for 3 training sets. The MSAF-RATIO-27-MULTIEXPANDED-4-GROUPS method computed 12 common frequency components for the electric impact drill: 355, 356, 371, 433, 707, 709, 865, 866, 1108, 1110, 1112, and 1297 Hz for *TCF* = 0.25 (6/24). The computed frequency components were used for feature vectors (see Tables 1–5 and Figures 18–22).

Table 1. Values of features of the vector **hd** (healthy electric impact drill).

Value of Feature			
0.002389	0.003904	0.000189	0.001208
0.006762	0.000414	0.000723	0.000510
0.002307	0.003273	0.006422	0.001074

Table 2. Values of features of the vector **ddgt** (electric impact drill with damaged gear train).

Value of Feature			
0.002216	0.002847	0.000434	0.001511
0.002599	0.000233	0.000449	0.000704
0.001381	0.001658	0.002525	0.003383

Table 3. Values of features of the vector **dfive** (electric impact drill with a faulty fan with 5 broken rotor blades).

Value of Feature			
0.001651	0.004912	0.012708	0.003004
0.006170	0.014284	0.002118	0.001150
0.005399	0.002338	0.004434	0.007732

Table 4. Values of features of the vector **dten** (electric impact drill with a faulty fan with 10 broken rotor blades).

Value of Feature			
0.003078	0.005273	0.000338	0.001864
0.006732	0.001138	0.001405	0.001365
0.001450	0.002512	0.007226	0.001697

Table 5. Values of features of the electric impact drill with a shifted brush (motor off).

Value of Feature			
0.0001033	0.0000653	0.0000749	0.0001034
0.0000923	0.0001318	0.0000371	0.0001065
0.0002000	0.0000667	0.0000323	0.0001145

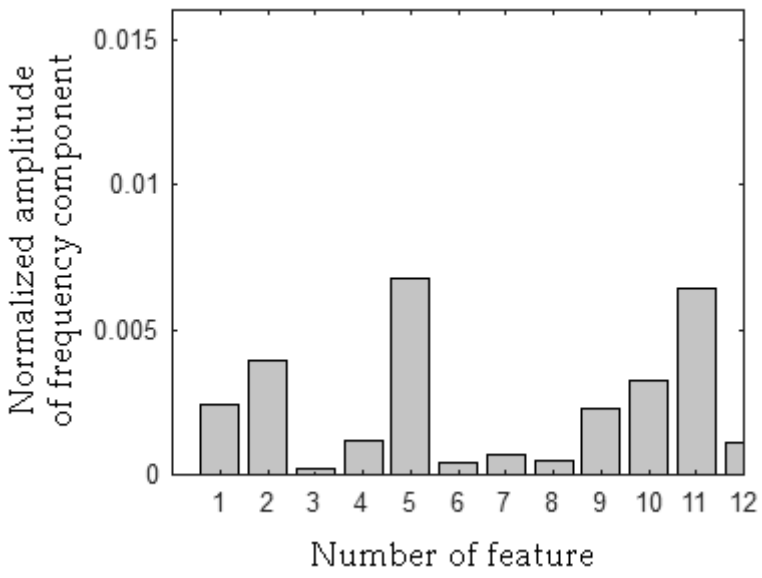


Figure 18. Values of features of the vector **hd** (healthy electric impact drill).

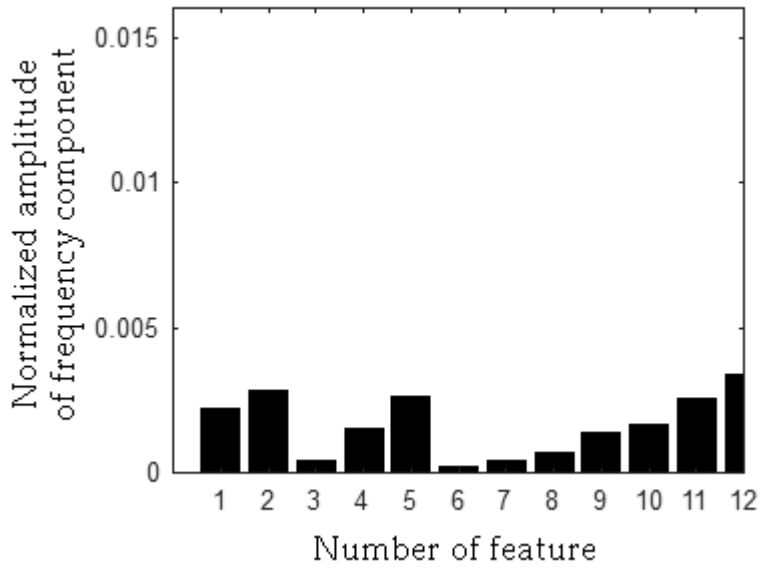


Figure 19. Values of features of the vector **ddgt** (electric impact drill with damaged gear train).

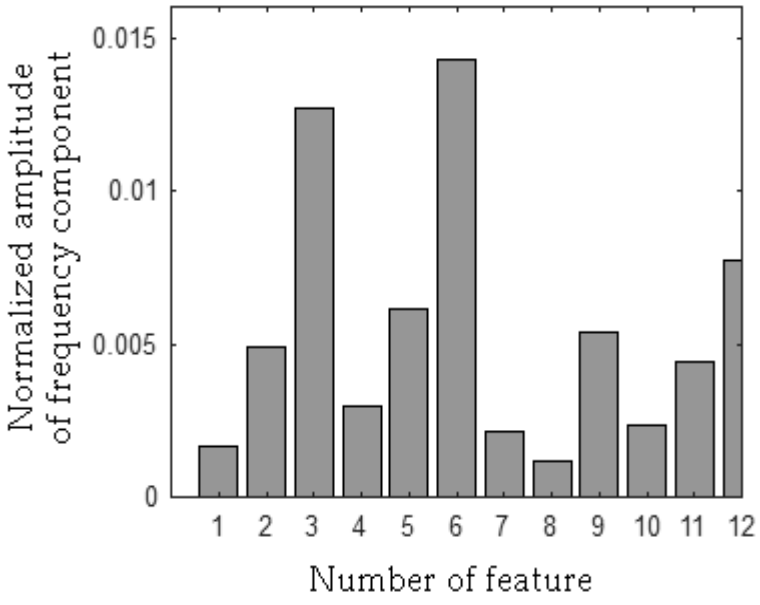


Figure 20. Values of features of the vector **dfive** (electric impact drill with a faulty fan with 5 broken rotor blades).

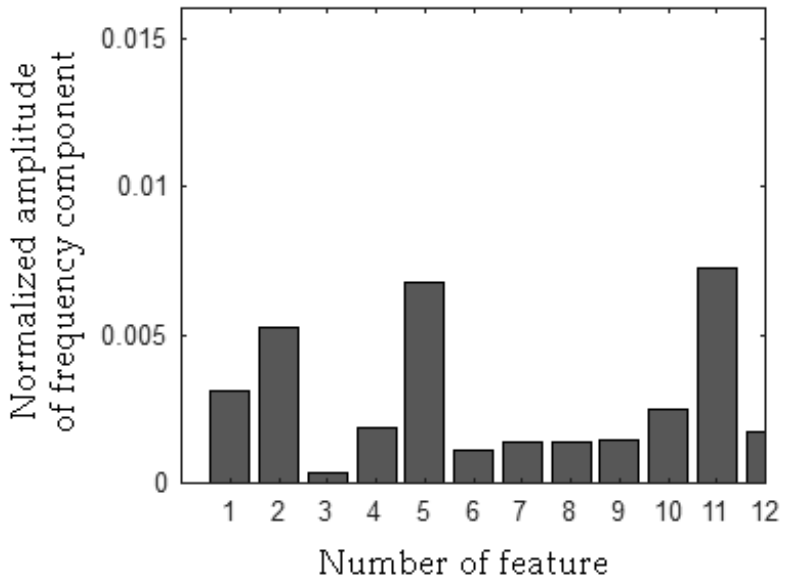


Figure 21. Values of features of the vector **dten** (electric impact drill with a faulty fan with 10 broken rotor blades).

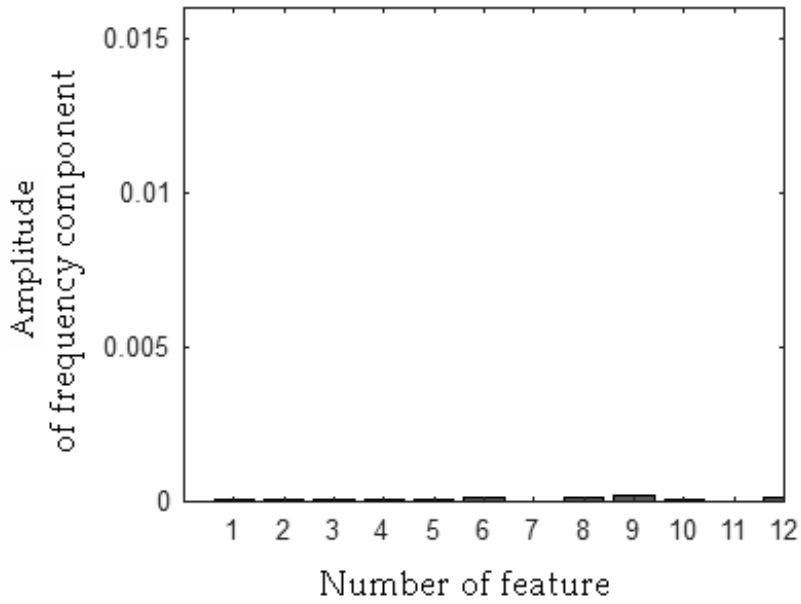


Figure 22. Values of features (without amplitude normalization) of the electric impact drill with a shifted brush (motor off).

The MSAF-RATIO-27-MULTIEXPANDED-4-GROUPS method computed 12 common frequency components of the blender: 280, 281, 292, 293, 295, 296, 479, 480, 1915, 1916, 2383, and 2479 Hz for *TCF* = 0.222 (4/18). The computed frequency components were used for feature vectors (Tables 6–8) (Figures 23–25).

Table 6. Values of features of the healthy blender.

Value of Feature			
0.002236667	0.012738667	0.001946333	0.003069333
0.020384	0.001906667	0.002930667	0.013355
0.001388333	0.003803	0.014756	0.002679333

Table 7. Values of features of the blender with a faulty fan (2 broken rotor blades).

Value of Feature			
0.000446667	0.000665333	0.014109667	0.000445333
0.000905333	0.0125	0.000423667	0.000836333
0.009938	0.000623333	0.000480667	0.008561667

Table 8. Values of features of the blender with a faulty fan (5 broken rotor blades).

Value of Feature			
0.136920667	0.000855333	0.000701667	0.059152333
0.005108	0.000530667	0.112844667	0.004174333
0.000560667	0.041357333	0.001391333	0.000543

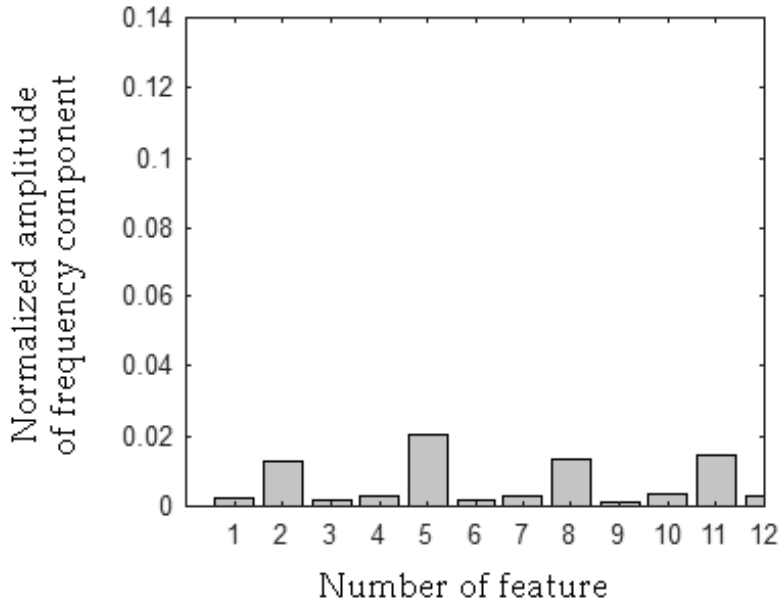


Figure 23. Values of features of the healthy blender.

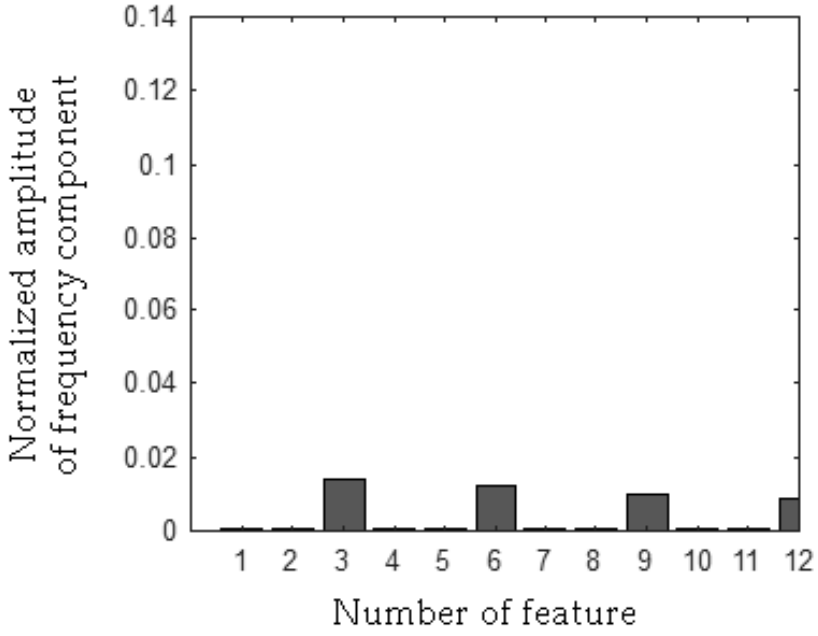


Figure 24. Values of features of the blender with (2 broken rotor blades).

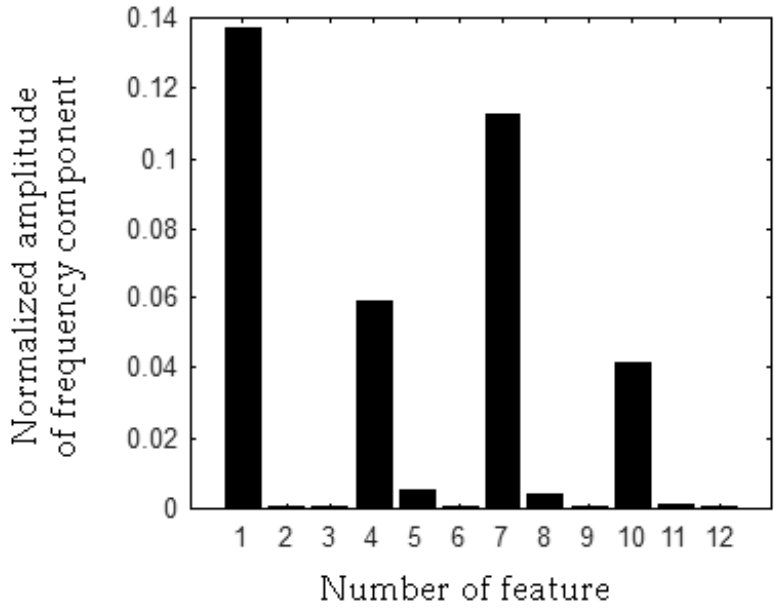


Figure 25. Values of features of the blender with a faulty fan (5 broken rotor blades).

Feature vectors were processed by the nearest mean (NM) classifier [29–36] and the SVM classifier [37] in the next step. The classification step used the NM classifier or SVM. However, neural networks [38–40] could alternatively be used for this task.

2.2. NM Classifier

To classify feature vectors, the NM classifier was used. It is a well-known classifier used for various applications (similar to the nearest neighbour classifier and k-means clustering) [29–36]. It used an averaged feature vector. The averaged feature vector was the arithmetic average of the training feature vectors of one class. For example, for 5 classes, there are 5 vectors. Next, the averaged feature vector was compared with the test feature vector using the Manhattan distance (Equation (2)).

$$MD(\mathbf{a} - \mathbf{hd}) = \sum_{i=1}^{1} |(a_i - hd_i)| \tag{2}$$

where $MD(\mathbf{a} - \mathbf{hd})$ is the Manhattan distance for vectors **a** and **hd**, where the test vector **a** = [a_{264}, a_{265}, a_{276}, a_{322}, a_{526}, a_{527}, a_{643}, a_{644}, a_{824}, a_{825}, a_{827}, a_{964}] and the averaged vector **hd** = [hd_{264}, hd_{265}, hd_{276}, hd_{322}, hd_{526}, hd_{527}, hd_{643}, hd_{644}, hd_{824}, hd_{825}, hd_{827}, hd_{964}].

The NM classifier performed Equation (2) for all feature vectors: $|\mathbf{a} - \mathbf{hd}|$, $|\mathbf{a} - \mathbf{ddgt}|$, $|\mathbf{a} - \mathbf{dfive}|$, $|\mathbf{a} - \mathbf{dten}|$. The nearest distance between vectors was essential for making decisions about the recognised class. For example, if $|\mathbf{a} - \mathbf{hd}|$ is the nearest distance, then the class "healthy electric impact drill" will be recognised.

There are 2 recordings of the electric impact drill in the presented approach. The first recording was split into training samples. The second recording was split into test samples. The MSAF-RATIO-27-MULTIEXPANDED-4-GROUPS method used the training samples. Next, it computed the common frequency components (in the analysed case, it was 12 common frequency components; please see Section 2). The number of common frequency components depended on the analysed signals. Equation (2) uses the unknown test vector "**a**" and the known training vector "**hd**". Each new recorded test sample is compared with the "training set" (all averaged feature vector). Assuming a signal for a new drill is acquired; next, it is compared with all states in the training set. Moreover, there is also the possibility of adding new training samples and new type of motors to the training set.

2.3. SVM Classifier

The second method of classification of the processed computed feature vectors (12 dimensions/features in the presented analysis) was using the SVM (support vector machine) classifier [37]. It was the supervised classification method. The SVM used a separating hyperplane. The method computed the separating hyperplane which categorised the unknown test feature vectors. Two classes of feature vectors could be separated by many hyperplanes. The objective of the SVM classifier was to compute the separating hyperplane that had the maximum margin [37]. The margin was the maximum distance between the analysed feature vectors. The hyperplane was a boundary between the analysed classes. It helped categorise the feature vectors. Feature vectors could be attributed to two classes (for example, healthy electric impact drill and electric impact drill with damaged gear train). This depended on the features of the vector and the location of the hyperplane.

The hyperplane was a line if the number of elements of feature vectors was equal to two. The hyperplane was a two-dimensional plane if the number of elements of feature vectors was equal to three.

Computed support vectors were the feature vectors. Support vectors were close to the hyperplane. The SVM model used support vectors to compute the position of the hyperplane. Distances between support vectors and the hyperplane were essential for the proper building of the SVM model (Figure 26).

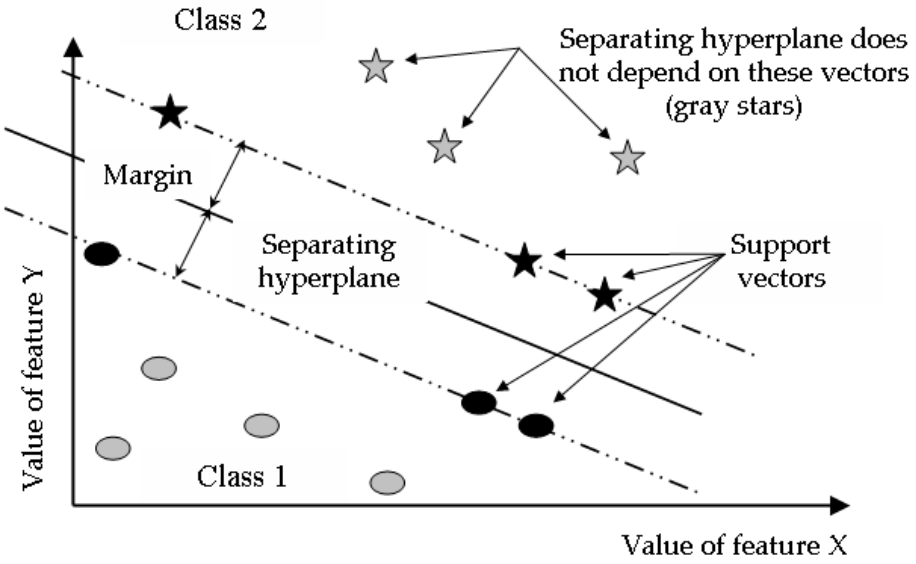


Figure 26. SVM model (for the example two features of X and Y).

3. Results of the Fault Detection of the Analysed Commutator Motors

Measurements of the acoustic signals of the analysed commutator motors (the commutator motor of the electric impact drill and the commutator motor of the blender) were conducted in a workshop. The analysed devices were operational, except for the electric impact drill with a shifted brush (motor off). The author measured and analysed five states of the electric impact drill: healthy, with damaged gear train, with a faulty fan with five broken rotor blades, with a faulty fan with 10 broken rotor blades, and with a shifted brush (motor off). Each electric impact drill had a commutator motor and had P_{EID} = 500 W, R_{EID} = 3000 rpm, V_{EID} = 230 V, f_{EID} = 50 Hz, and M_{EID} = 1.84 kg, where M_{EID}—weight of the electric impact drill, P_{EID}—rated power of the electric impact drill, R_{EID}—rotation speed of the electric impact drill, V_{EID}—supply voltage of the electric impact drill, and f_{EID}—current frequency of the electric impact drill. The author measured and analysed three states of the blender: healthy, with a faulty fan with two broken rotor blades, and with a faulty fan with 5 broken rotor blades. Each blender had a commutator motor and had P_B = 300 W, R_B = 10,000 rpm, V_B = 230 V,

and f_B = 50 Hz, M_B = 1.75 kg, where M_B—weight of the blender, P_B—rated power of the blender, R_B—rotation speed of the blender, V_B—supply voltage of the blender, and f_B—current frequency of the blender.

To compute patterns, 20 one-second training samples of the electric impact drill and 18 one-second training samples of the blender were analysed. To analyse the proposed approach (please see Section 2), 125 one-second test samples of the electric impact drill and 102 one-second test samples of the blender were analysed. Acoustic data of the analysed commutator motors were evaluated by Equation (3). It defined the efficiency of recognition of the electric impact drill/blender (E_{EID}):

$$E_{EID} = (N_{EID})/(N_{AEID}) \times 100\% \quad (3)$$

where E_{EID}—the efficiency of recognition of the electric impact drill/blender for the selected type of acoustic signal, N_{EID}—the number of test samples of the electric impact drill/blender tested properly for the selected type of acoustic signal, and N_{AEID}—the number of all test samples of the electric impact drill/blender for the selected type of acoustic signal.

The total efficiency of recognition of the electric impact drill (TE_{EID}) was used to evaluate the E_{EID} of all analysed acoustic signals of the electric impact drill. TE_{EID} was expressed as Equation (4):

$$TE_{EID} = (E_{EID1} + E_{EID2} + E_{EID3} + E_{EID4} + E_{EID5})/5 \quad (4)$$

where TE_{EID}—the total efficiency of recognition of the electric impact drill for the five selected types of acoustic signals, E_{EID1}—the efficiency of recognition of the healthy electric impact drill, E_{EID2}—the efficiency of recognition of the electric impact drill with a damaged gear train, E_{EID3}—the efficiency of recognition of the electric impact drill with a faulty fan (five broken rotor blades), E_{EID4}—the efficiency of recognition of the electric impact drill with a faulty fan (10 broken rotor blades), and E_{EID5}—the efficiency of recognition of the electric impact drill with a shifted brush (motor off).

The total efficiency of recognition of the blender (TE_B) was expressed as Equation (5):

$$TE_B = (E_{EID-B1} + E_{EID-B2} + E_{EID-B3})/3 \quad (5)$$

where TE_B—the total efficiency of recognition of the blender for three selected types of acoustic signals, E_{EID-B1}—the efficiency of recognition of the healthy blender, E_{EID-B2}—the efficiency of recognition of the blender with a faulty fan (two broken rotor blades), E_{EID-B3}—the efficiency of recognition of the blender with a faulty fan (five broken rotor blades).

Computed TE_{EID} and E_{EID} values are shown in Tables 9 and 10. The analysis of the acoustic signals of the electric impact drill using the MSAF-RATIO-27-MULTIEXPANDED-4-GROUPS method (12 analysed frequency components) and the NM was carried out (see Table 9).

Table 9. Analysis of the acoustic signals of the electric impact drill using the MSAF-RATIO-27-MULTIEXPANDED-4-GROUPS method (12 analysed frequency components) and the NM.

Type of Acoustic Signal	E_{EID} [%]
Healthy electric impact drill	100
Electric impact drill with a damaged gear train	100
Electric impact drill with a faulty fan (5 broken rotor blades)	84
Electric impact drill with a faulty fan (10 broken rotor blades)	96
Electric impact drill with a shifted brush (motor off)	100
	TE_{EID} [%]
Total efficiency of recognition of the electric impact drill	96

Table 10. Analysis of the acoustic signals of the electric impact drill using the MSAF-RATIO-27-MULTIEXPANDED-4-GROUPS method (12 analysed frequency components) and SVM.

Type of Acoustic Signal	E_{EID} [%]
Healthy electric impact drill	96
Electric impact drill with a damaged gear train	100
Electric impact drill with a faulty fan (5 broken rotor blades)	76
Electric impact drill with a faulty fan (10 broken rotor blades)	72
Electric impact drill with a shifted brush (motor off)	100
	TE_{EID} [%]
Total efficiency of recognition of the electric impact drill	88.8

The analysis of the acoustic signals of the electric impact drill using the MSAF-RATIO-27-MULTIEXPANDED-4-GROUPS method (12 analysed frequency components) and SVM was also carried out (see Table 10).

The computed results of the analysis of the electric impact drill were good (E_{EID} = 84–100% and TE_{EID} = 96%) for the NM classifier. The computed results of the analysis of the electric impact drill were the following: E_{EID} = 72–100% and TE_{EID} = 88.8% for the SVM classifier. The MSAF-RATIO-27-MULTIEXPANDED-4-GROUPS method used 12 frequency components of the acoustic signals of the electric impact drill.

The analysis of the acoustic signals of the blender using the MSAF-RATIO-27-MULTIEXPANDED-4-GROUPS method (12 analysed frequency components) and the NM was carried out (see Table 11).

Table 11. Analysis of the acoustic signals of the blender using the MSAF-RATIO-27-MULTIEXPANDED-4-GROUPS method (12 analysed frequency components) and the NM.

Type of Acoustic Signal	E_{EID} [%]
Healthy blender	100
Blender with a faulty fan (2 broken rotor blades)	100
Blender with a faulty fan (5 broken rotor blades)	100
	TE_B [%]
Total efficiency of recognition of the blender	100

The analysis of the acoustic signals of the blender using the MSAF-RATIO-27-MULTIEXPANDED-4-GROUPS method (12 analysed frequency components) and SVM was also carried out (Table 12).

Table 12. Analysis of the acoustic signals of the blender using the MSAF-RATIO-27-MULTIEXPANDED-4-GROUPS method (12 analysed frequency components) and SVM.

Type of Acoustic Signal	E_{EID} [%]
Healthy blender	82
Blender with a faulty fan (2 broken rotor blades)	100
Blender with a faulty fan (5 broken rotor blades)	100
	TE_B [%]
Total efficiency of recognition of the blender	94.11

The computed results of the analysis of the blender were good (E_{EID} = 100% and TE_B = 100%) for the NM classifier. The computed results of the analysis of the blender were the following: E_{EID} = 82–100% and TE_B = 94.11% for the SVM classifier. The MSAF-RATIO-27-MULTIEXPANDED-4-GROUPS method used 12 frequency components of the acoustic signals of the blender.

4. Discussion

The dependency of performance on the acoustic sensor was significant. The results of recognition depended on the training samples. If the training samples are measured by a specific type of microphone, then new test samples should be also measured by the same type of microphone. Frequency components can be different for different microphones, but the results should be similar.

Another point is the detection of a new, unknown fault. If there is an unknown fault of the commutator motor, then it will be classified as the nearest neighbour (the nearest spectrum of frequency). For example, if there is an acoustic signal of an aircraft, it will be classified as unknown (for example, the class 'unknown' includes the sounds of cars, trains, and aircrafts).

The proposed acoustic-based fault detection with variant selection of methods of recognition was based on the difference of FFT spectra (with normalized amplitude). The analysed differences were not big. The results of the recognition of acoustic signals (see Tables 9–12) showed that the NM classifier was better than the SVM classifier. The nearest neighbour classifier and the nearest mean classifier are based on distance metrics. Distance metrics were very good for the classification of nonlinearly separable data and linearly separable data. In this paper, the SVM classifier used a linear function, the LSVM (linear SVM), which can classify linearly separable data.

5. Summary and Conclusions

The article presented a means of acoustic-based fault detection with variant selection of methods of recognition for two commutator motors: the commutator motor of an electric impact drill and the commutator motor of a blender. Five states of the electric impact drill were analysed: healthy, with a damaged gear train, with a faulty fan with 5 broken rotor blades, with a faulty fan with 10 broken rotor blades, and with a shifted brush (motor off). Three states of the blender were analysed: healthy, with a faulty fan with 2 broken rotor blades, and with a faulty fan with 5 broken rotor blades.

The MSAF-RATIO-27-MULTIEXPANDED-4-GROUPS method was used for the computation of feature vectors. The NM and SVM classifiers were used for data classification.

The computed results of analysis of the electric impact drill were good, with E_{EID} = 84–100% and TE_{EID} = 96% for the NM classifier. The computed results of analysis of the electric impact drill were E_{EID} = 72–100% and TE_{EID} = 88.8% for the SVM classifier. The computed results of analysis of the blender were very good, with E_{EID} = 100% and TE_B = 100% for the NM classifier. The computed results of analysis of the blender were E_{EID} = 82–100% and TE_B = 94.11% for the SVM classifier.

The proposed approach did not have a high cost. An ordinary computer and an ordinary smartphone cost about $600. The proposed solution can be used in other applications, such as the fault detection of angle grinders, cordless drills, hair dryers, various types of rotating machines, etc. The proposed approach can find applications in the energy industry, material industry, construction industry, mining industry, transport industry, and for construction and renovation companies. However, there were disadvantages of the proposed analysis: the appearance of background noises and reflected sounds.

The proposed approach can be developed in the future. Acoustic signals of various types of rotating machines can be measured and analysed. Acoustic signals can be measured by many types of microphones or acoustic cameras. There is also the possibility of adding diagnostic signals, such as vibrations or thermal images. To achieve a higher recognition efficiency, new feature extraction and classification methods can be analysed.

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